Name: _____ Date: 22/09/2017

Array lists with objects

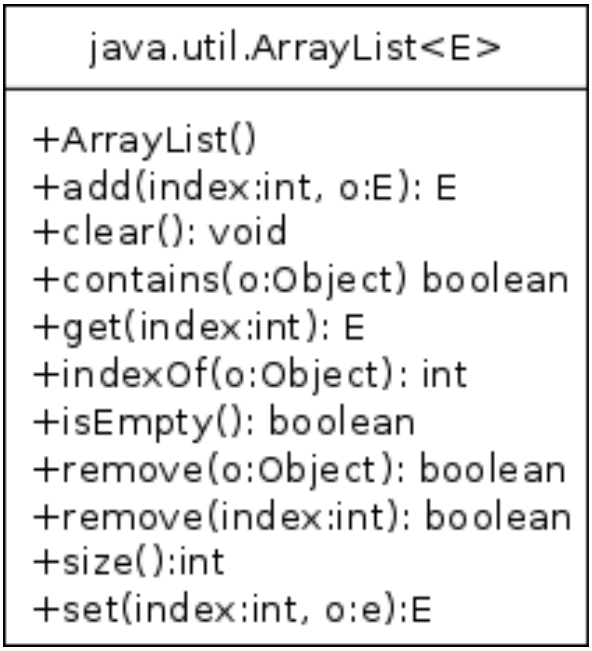
This program provided to you is a skeleton of a small system to keep track of teachers and their information.

You will write the code needed to complete the following tasks:

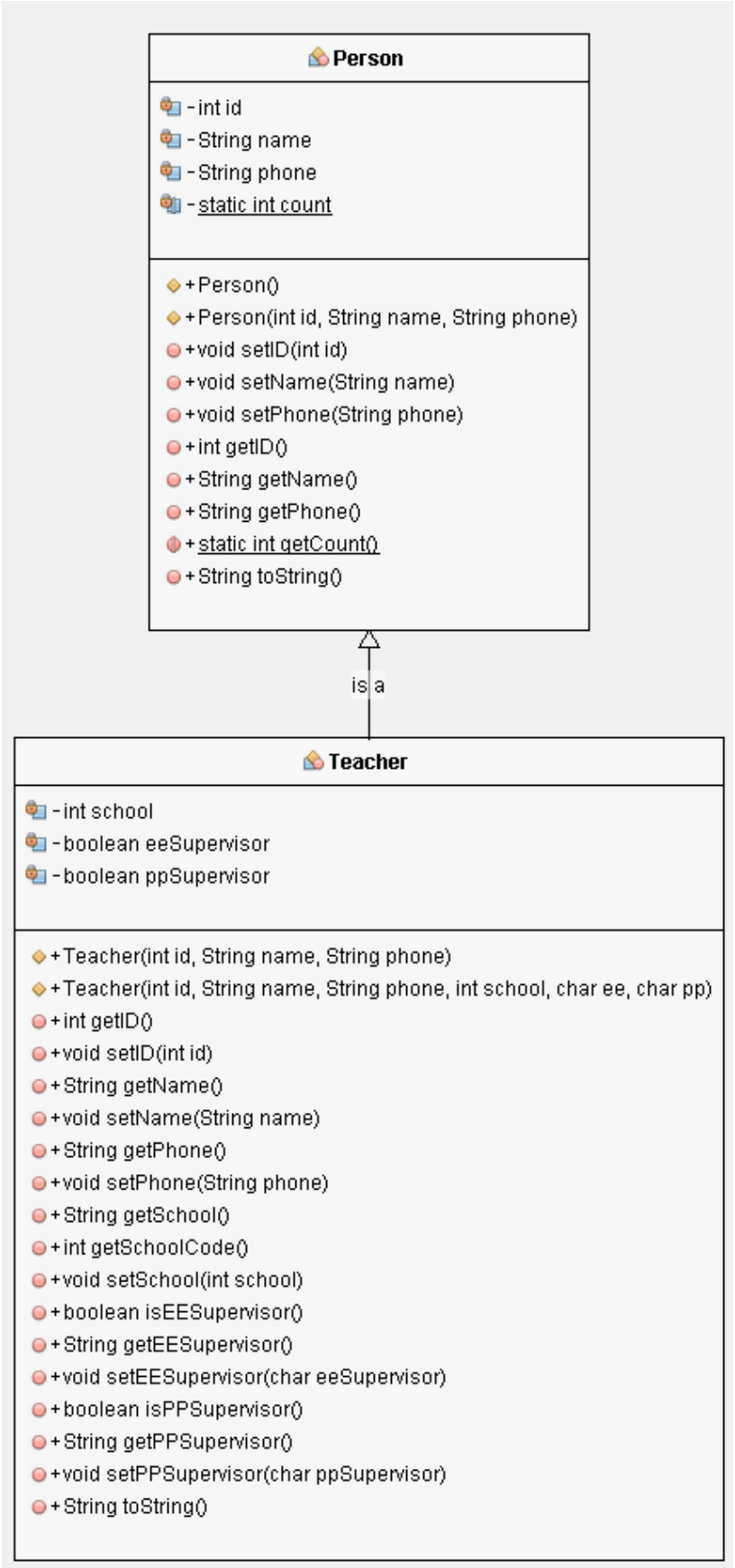
1. Validate the input of school code in the main menu (1 ≤ code ≤ 4)
2. Validate the input of an id in the main menu (id > 0, unique)
3. Produce a full list containing all teacher data for all teachers
4. Produce a compact list containing only id, name and school of each teacher
5. Create an array list of teachers of a specific school
6. Create an array list of teachers supervising extended essays
7. Create an array list of teachers supervising personal projects
8. Create a list with results of searching teachers by a name or part of a name
9. Search a teacher given their id number
10. Create an array list of teachers supervising both EE and PP
11. Edit a teacher record (edit teacher information, delete the record, cancel)

Work through the test from the beginning. Your program should build and grow –do not start a new program for each point. During this test you may use **any** resources that you have created, class resources and notes but you may **not** use Internet.

Refer to the class diagrams given to write your code.



<<< Please Turn Over >>>


Person
- int id
- String name
- String phone
- static int count
+ Person()
+ Person(int id, String name, String phone)
+ void setID(int id)
+ void setName(String name)
+ void setPhone(String phone)
+ int getID()
+ String getName()
+ String getPhone()
+ static int getCount()
+ String toString()
is a
Teacher
- int school
- boolean eeSupervisor
- boolean ppSupervisor
+ Teacher(int id, String name, String phone)
+ Teacher(int id, String name, String phone, int school, char ee, char pp)
+ int getID()
+ void setID(int id)
+ String getName()
+ void setName(String name)
+ String getPhone()
+ void setPhone(String phone)
+ String getSchool()
+ int getSchoolCode()
+ void setSchool(int school)
+ boolean isEESupervisor()
+ String getEESupervisor()
+ void setEESupervisor(char eeSupervisor)
+ boolean isPPSupervisor()
+ String getPPSupervisor()
+ void setPPSupervisor(char ppSupervisor)
+ String toString()

Instructions	***Program Display/Screen Output***

1. Validate the input of school code in the main menu	`Enter one letter for your option: s`
	`Enter teacher school: [1-Kindy | 2-ES | 3-MS | 4-HS] 0`
	`Error - school code out of range.`
	`Enter teacher school: 5`
	`Error - school code out of range.`

2. Validate the input of an id in the main menu (id > 0)	`Enter one letter for your option: I`
	`Enter a teacher's id number to search: -12`
	`Error - id must be positive and > 0.`
	`Enter a teacher's id number to search: 99`
	`ID not found.`

3. Produce a full list containing all data for all teachers	`Enter one letter for your option: l`
	`Teacher ID : 1`
	`Teacher name : Albert`
	`Phone number : +01 1324 9873`
	`School : HS`
	`Supervising EE: Yes`
	`Supervising PP: No`
	`…`
	`Teacher ID : 12`
	`Teacher name : Delete`
	`Phone number :`
	`School : MS`
	`Supervising EE: No`
	`Supervising PP: No`

4. Produce a compact list containing only ***id***, ***name*** and ***school*** of each teacher	`Enter one letter for your option: c`
	`1 Albert HS`
	`...`
	`12 Delete MS`

5. Create an array list of teachers of a *specific school*	`Enter one letter for your option: s`
	`Enter teacher school: [1-Kindy | 2-ES | 3-MS | 4-HS] 2`
	`6 John ES`
	`7 Cole ES`

6. Create an array list of teachers supervising *Extended Essays* and another for those supervising *Personal Projects*	`Enter one letter for your option: e`
	`1 Albert HS`
	`2 Alan HS`
	`3 Bruce HS`
	`8 Kevin HS`
	`Enter one letter for your option: p`
	`2 Alan HS`
	`8 Kevin HS`
	`9 Clara MS`
	`10 Cain MS`

7. Create an array list of teachers supervising **both** *Extended Essays* and *Personal Projects*	`Enter one letter for your option: b`
	`2 Alan HS`
	`8 Kevin HS`

8. Create a list with results of searching teachers by a full or partial *name*	`Enter one letter for your option: n`
	`Enter name to search for: c`
	`7 Cole ES`
	`9 Clara MS`
	`10 Cain MS`

9. Look for a teacher given their *id* number	`Enter one letter for your option: i`
	`Enter a teacher's id number to search: 12`
	`Teacher ID : 12`
	`Teacher name : Delete`
	`Phone number :`
	`School : MS`
	`Supervising EE: No`
	`Supervising PP: No`

10. After finding the teacher, edit their record. Give options to edit teacher information, delete the record and cancel/abort the operation)	`[E]dit [D]elete [C]ancel:e`
	`Enter correct name : Deleteme`
	`Enter correct phone :`
	`Enter correct school: [1-Kindy | 2-ES | 3-MS | 4-HS] 1`
	`Supervising EE (y/n)? n`
	`Supervising PP (y/n) ? n`
	`Edited record:`
	`Teacher ID : 12`
	`Teacher name : Deleteme`
	`Phone number :`
	`School : Kindy`
	`Supervising EE: No`
	`Supervising PP: No`